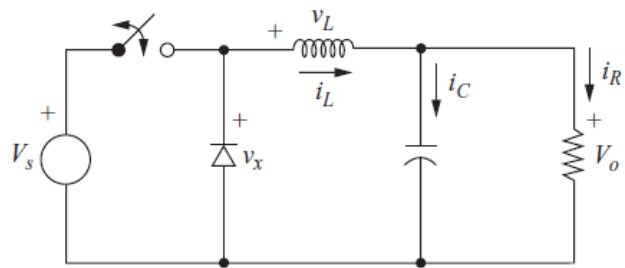
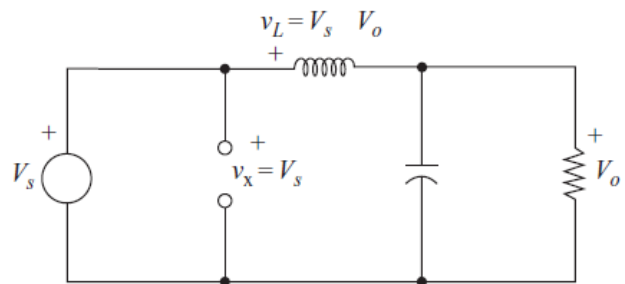


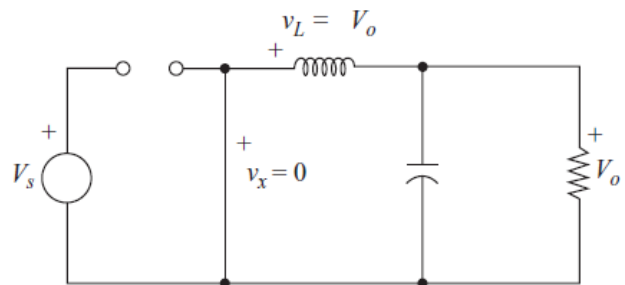
Modelagem do Conversor CC-CC Buck e Diagrama de Blocos do Controle



(a)

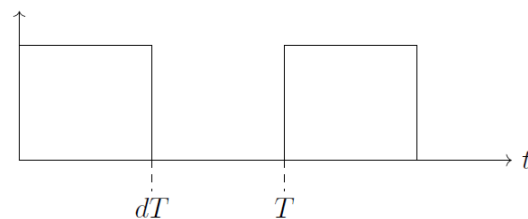
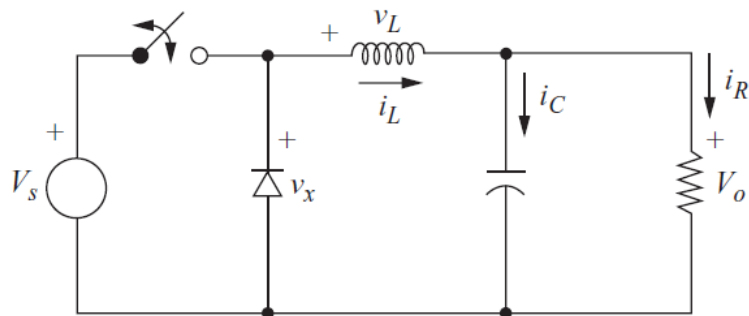


(b)



(c)

Obtenção da Função de Transferência dinâmica do Conversor Buck:



$$i_L(t) = i_C(t) + i_o(t)$$

$$i_o(t) = \frac{v_o(t)}{R}$$

$$i_C(t) = C \frac{dv_o(t)}{dt}$$

$$v_L(t) = L \frac{di_L(t)}{dt} = V_i d(t) - v_o(t)$$

$$V_i d(t) - v_o(t) = L \frac{d}{dt} \left(C \frac{dv_o(t)}{dt} + \frac{v_o(t)}{R} \right)$$

$$V_i d(t) - v_o(t) = LC \frac{d^2 v_o(t)}{dt^2} + \frac{L}{R} \frac{dv_o(t)}{dt}$$

$$V_i d(t) = LC \frac{d^2 v_o(t)}{dt^2} + \frac{L}{R} \frac{dv_o(t)}{dt} + v_o(t)$$

Linearização num ponto de operação:

$$d(t) = D + \tilde{d}(t)$$

$$v_o(t) = V_o + \tilde{v}_o(t)$$

$$V_i(D + \tilde{d}(t)) = LC \frac{d^2}{dt^2} (V_o + \tilde{v}_o(t)) + \frac{L}{R} \frac{d}{dt} (V_o + \tilde{v}_o(t)) + V_o + \tilde{v}_o(t)$$

$$V_i D + V_i \tilde{d}(t) = LC \frac{d^2 \tilde{v}_o(t)}{dt^2} + \frac{L}{R} \frac{d \tilde{v}_o(t)}{dt} + V_o + \tilde{v}_o(t)$$

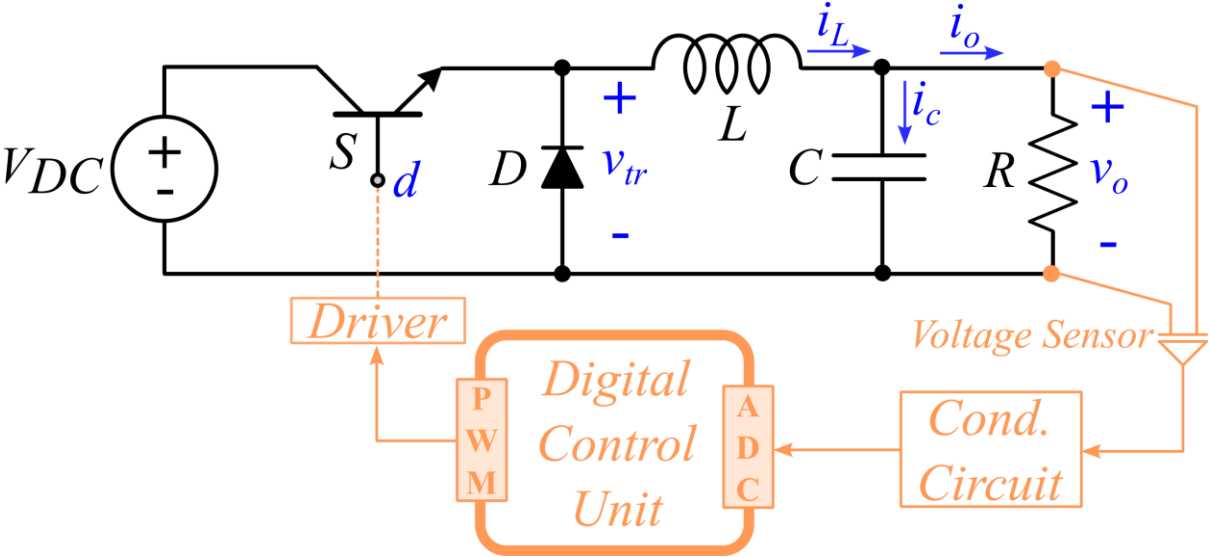
$$V_i D = V_o$$

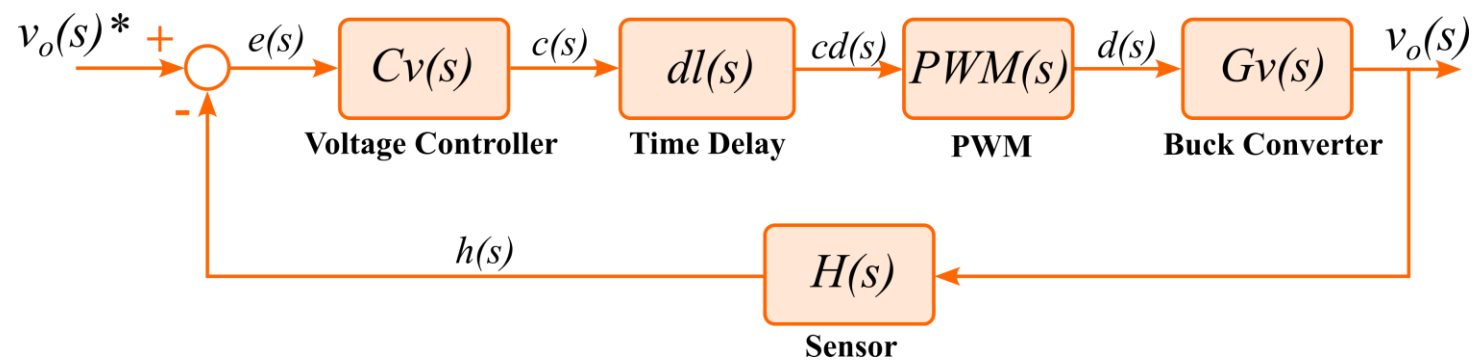
$$V_i \tilde{d}(t) = LC \frac{d^2 \tilde{v}_o(t)}{dt^2} + \frac{L}{R} \frac{d\tilde{v}_o(t)}{dt} + \tilde{v}_o(t)$$

$$V_i \tilde{d}(s) = \tilde{v}_o(s) \left(LC s^2 + \frac{L}{R} s + 1 \right) .$$

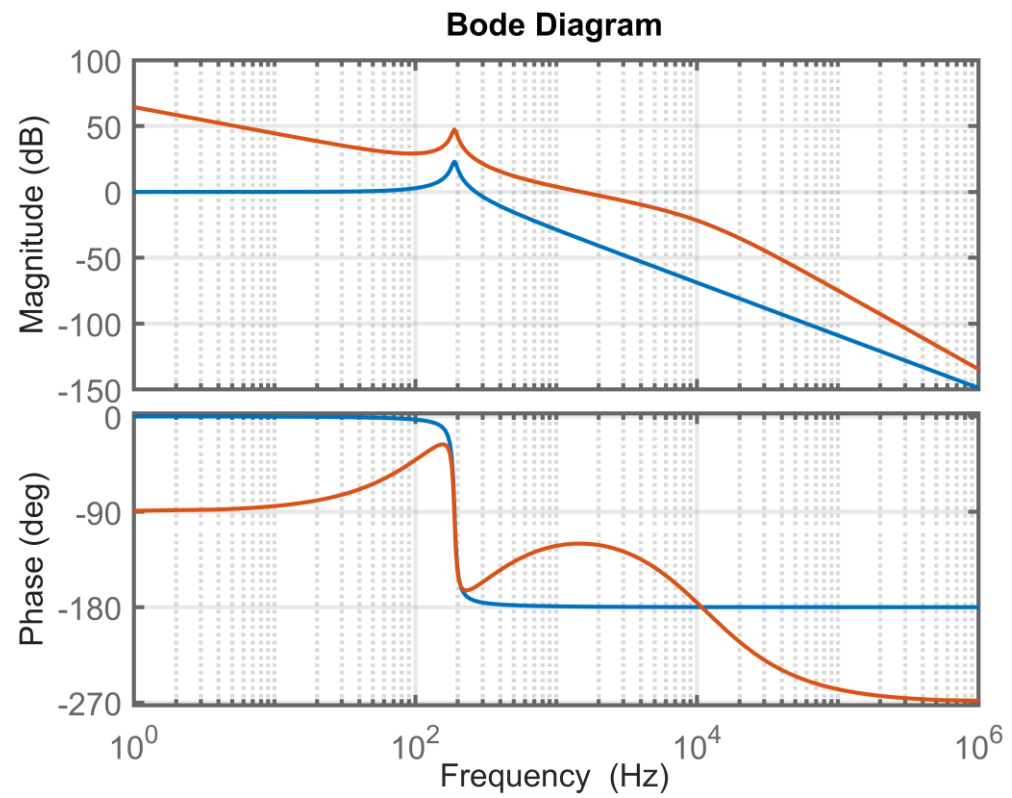
$\frac{\tilde{v}_o(s)}{\tilde{d}(s)} = \frac{V_i}{LC s^2 + \frac{L}{R} s + 1} = G_v(s)$

Diagrama de Blocos:

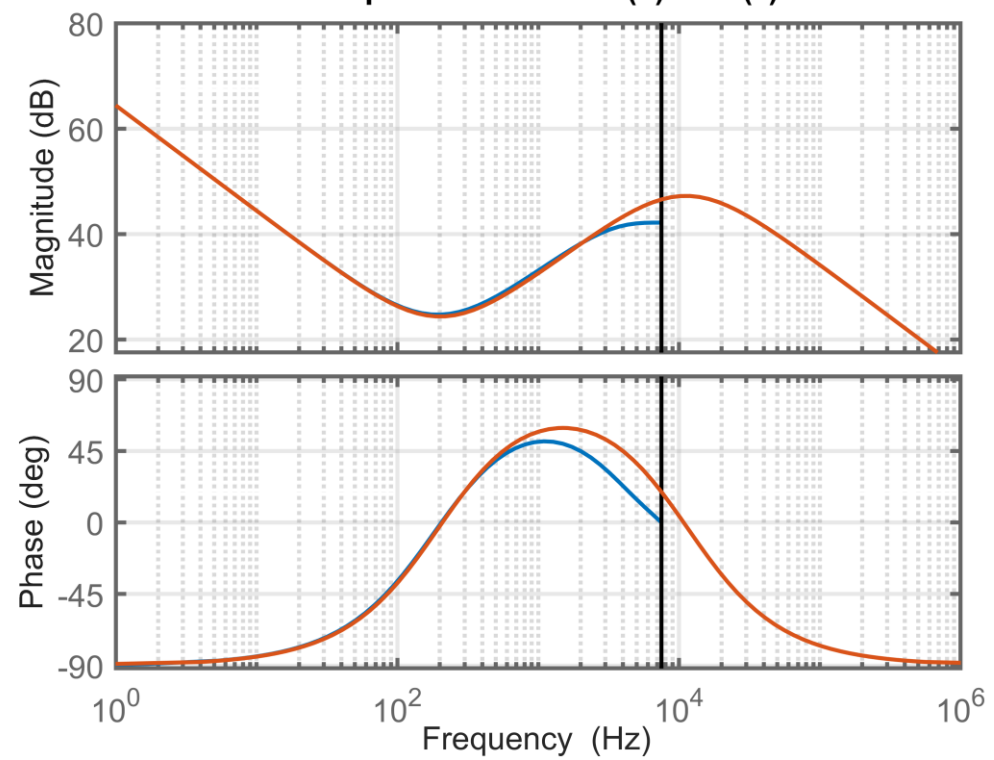




Exemplo de diagramas de bode de FTMAu(s) e FTMAc(s):



Comparativo entre $C_v(s)$ e $C_v(z)$



Bode em Malha fechada

